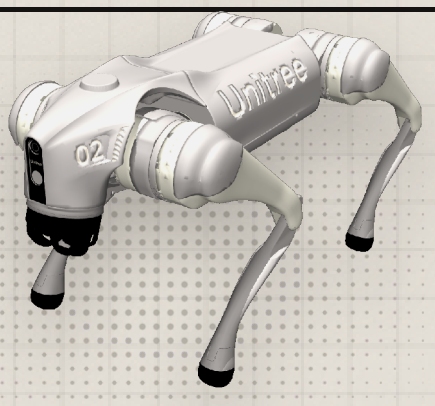
A robot dog that checks if the route is passable.

A Unitree Go2, **driven by a person with the remote**, records camera, LiDAR and pose.



UNITREE GO2 · OFFICIAL MODEL RENDER

“I got injured and ended up in a wheelchair … I had no idea if my route to the bus was even passable.”

SHIVRAJ, WHO BUILT BENBEN

TODAY, OWNERS

learn about barriers from whoever couldn’t pass.

TODAY, ROUTING APPS

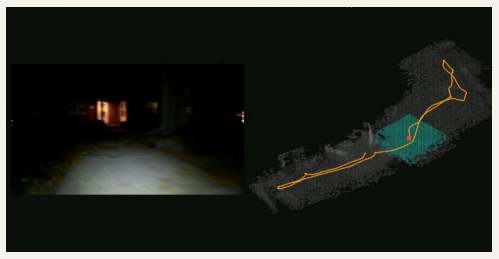
run on map data that can be years old.

01 WHAT WE BUILT



01 Drive

A person pilots it by remote.



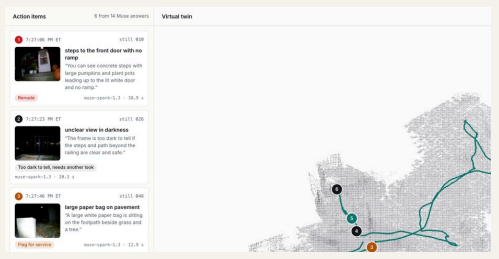
02 Sense

A listen-only link records it all.



03 Judge

Muse Spark picks one of four actions. SAM 3.1 boxes the obstacle.



04 Act

Each finding keeps its photo and capture time.

02 WHAT RAN ON NOV 4

RECORDED

3,396 video frames

2,355 LiDAR clouds
5 CM VOXELS · ~10 HZ

227 timestamped stills
POSE AT ~19 HZ

14 stills judged
BY MUSE SPARK

~4 MIN · AFTER DARK · HUMAN-PILOTED

MUSE SPARK VERDICTS · MUSE-SPARK-1.3 — 14 STILLS



- **1** reroute · steps, no ramp
- **2** flag · paper bag
- **1** report · entrance step
- **10** needs a human
- **8** of them too dark

MUSE SPARK
7–43 s / frame
One action per still, with a reason.

SAM 3.1 · LIVE
0.8–2.4 s / frame
Boxed the stairs, bag, chair and entrance step.

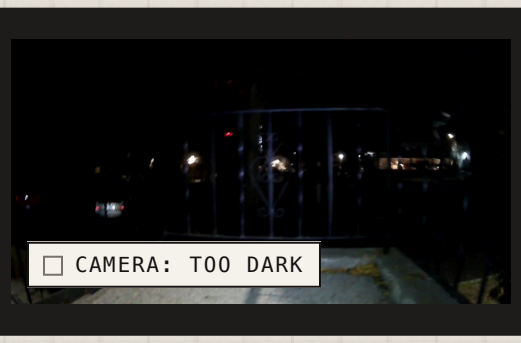
INDOOR TWIN · M1 PRO
4.2M points in **122 s**
MAPANYTHING · 32 FRAMES
BRUSH SPLAT · 17 DB PSNR, HELD-OUT VIEWS

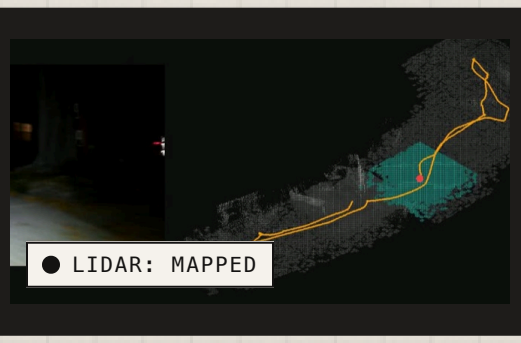
HOW IT FITS

- ● Unitree Go2 — REMOTE-PILOTED
- ● Listen-only recorder — WEBRTC · MAC
- ● Run folder — VIDEO · LIDAR · POSE
- ● Muse Spark · ● SAM 3.1
- ● Workbench — RAN LOCALLY
- ◐ Ask-Muse · ◐ Voice
- ○ Place memory · ○ Route patrol

No motion commands sent. A person drove BenBen for the whole run.

03 WHAT WE LEARNED





8 of 14 verdicts: too dark. The LiDAR mapped the same stretches.

Geometry decides when to look. Vision decides what it is.

04 NEXT 90 DAYS

○ 01
Tier-0 gate
BLUR · DARKNESS · LIDAR HEIGHT-MAP DIFF, BEFORE MUSE

○ 02
Checkpoint memory
KEYED BY ROUTE CHECKPOINT, NOT OBJECT IDENTITY

○ 03
Append-only log
STATE = LATEST USABLE OBSERVATION

○ 04
One targeted question
“NEEDS A HUMAN” → ONE QUESTION + SAM BOX

○ 05
Clear, blocked, cleared
STAGED DAYLIGHT PAIR: REVISIT + VERIFY

Meta stack Muse Spark · SAM 3.1 · MapAnything · Voice Transcribe · `github.com/shivraj-S-bhatti/benben`

● ran Nov 4 ◐ wired, tested offline ○ roadmap